

R version 4.6.0 (2026-04-24 ucrt) -- "Because it was There"
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 Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
 You are welcome to redistribute it under certain conditions.
 Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
 Type 'contributors()' for more information and
 'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
 'help.start()' for an HTML browser interface to help.
 Type 'q()' to quit R.

[Previously saved workspace restored]

```
> library(pdftools)
Using poppler version 26.01.0
> library(readxl)
> library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
> library(stringr)
> library(writexl)
>
> # =====
> # PATH
> # =====
>
> mdna_folder <- "D:/SKRIPSI/output_MD&A"
>
> dictionary_file <- "D:/SKRIPSI/Loughran-McDonald_MasterDictionary_1993-2025.xlsx"
>
> output_file <- "D:/SKRIPSI/TONE.xlsx"
>
> # =====
> # BACA KAMUS LOUGHRAN-MCDONALD
> # =====
>
> dict <- read_excel(dictionary_file)
-|
>
> positive_words <- dict$Word[dict$Positive > 0]
> negative_words <- dict$Word[dict$Negative > 0]
>
> positive_words <- toupper(positive_words)
> negative_words <- toupper(negative_words)
>
> cat("Jumlah kata positif :", length(positive_words), "\n")
Jumlah kata positif : 347
> cat("Jumlah kata negatif :", length(negative_words), "\n")
Jumlah kata negatif : 2345
>
```

```

> # =====
> # DAFTAR FILE PDF MD&A
> # =====
>
> pdf_files <- list.files(
+   path = mdna_folder,
+   pattern = "\\\\.pdf$",
+   full.names = TRUE,
+   ignore.case = TRUE
+ )
>
> cat("Jumlah file MD&A :", length(pdf_files), "\n")
Jumlah file MD&A : 328
>
> # =====
> # LOOPING HITUNG TONE
> # =====
>
> hasil <- list()
>
> for(i in seq_along(pdf_files)) {
+
+   file_path <- pdf_files[i]
+
+   file_name <- tools::file_path_sans_ext(
+     basename(file_path)
+   )
+
+   cat("Processing :", file_name, "\n")
+
+   tryCatch({
+
+     text <- pdf_text(file_path)
+
+     text <- paste(text, collapse = " ")
+
+     text <- toupper(text)
+
+     text <- gsub("[^A-Z ]", " ", text)
+
+     words <- unlist(str_split(text, "\\s+"))
+
+     words <- words[words != ""]
+
+     positive_count <- sum(words %in% positive_words)
+
+     negative_count <- sum(words %in% negative_words)
+
+     total_words <- length(words)
+
+     tone <- ifelse(
+       positive_count + negative_count == 0,
+       NA,
+       (positive_count - negative_count) /
+       (positive_count + negative_count)
+     )
+
+     split_name <- str_split(
+       file_name,
+       "_",
+       simplify = TRUE
+     )
+
+     firm_code <- split_name[1]
+
+     year <- as.numeric(split_name[2])
+
+     hasil[[i]] <- data.frame(

```

```
+      Firm_Code = firm_code,  
+      Year = year,  
+      Positive = positive_count,  
+      Negative = negative_count,  
+      Total_Words = total_words,  
+      TONE = tone  
+    )  
+  
+  }, error = function(e){  
+  
+    cat("ERROR :", file_name, "\n")  
+  
+  })  
+ }  
+ }
```

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Processing : AALI_2012_MD&A

Processing : AALI_2013_MD&A

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Processing : UNVR_2022_MD&A
>
> # =====
> # GABUNG HASIL
> # =====
>
> tone_data <- bind_rows(hasil)
>
> tone_data <- tone_data %>%
+   arrange(Firm_Code, Year)
>
```

```
> # =====
> # EXPORT EXCEL
> # =====
>
> write_xlsx(
+   tone_data,
+   output_file
+ )
>
> # =====
> # RINGKASAN
> # =====
>
> cat("\n=====\\n")

=====
> cat("SELESAI\\n")
SELESAI
> cat("Jumlah observasi :", nrow(tone_data), "\\n")
Jumlah observasi : 328
> cat("Output :", output_file, "\\n")
Output : D:/SKRIPSI/TONE.xlsx
> cat("=====\\n")
```